

Education

- 2018–2022 **The Chinese University of Hong Kong**,
(Expected) **Ph.D. in Computer Science and Engineering**,
Supervisor: *Prof. James Cheng*.
- 2013–2017 **South China University of Technology**,
B.Eng in Computer Science and Technology.

Research Interests

Distributed system, machine learning framework, cluster scheduling.

Internships

- Oct 2021 - **Amazon Web Service, AI Lab**.
Now
 - Applied Scientist Intern in DGL Team
 - Distributed sampling on GPUs for GNN training
- Nov 2020 - **Alibaba Group, Apsara Platform**.
Sept 2021
 - Research Intern in Fuxi Team
 - Resource management for clusters with thousands of GPUs
- July 2019 - **HUAWEI, 2012 Lab**.
Sept 2020
 - Research Intern in Mindspore Team
 - Auto-parallelism systems for training large-scale DNN models
 - Efficient distributed GNN training systems
- July 2018 - **Alibaba Group, Apsara Platform**.
Aug 2018
 - Research Intern in MaxCompute Team
 - Smart index selection for OLAP on big tables in data warehouses
- July 2017 - **The Chinese University of Hong Kong**.
July 2018
 - Research Assistant supervised by **Prof. James Cheng**
 - Distributed Systems for large-scale data analytics

Awards

- 2018 - 2022 **Postgraduate Studentship, CUHK**.
- 2014 - 2016 **Undergraduate National Scholarship (three times), SCUT**.
- 2014 - 2016 **Merit Student, SCUT**.
- 2017 **ACM-ICPC World Finalist (Rank 56)**.
- 2016 **ACM-ICPC Regional Contest, Gold Medals**.
- 2016 **Guangdong Collegiate Programming Contest (GDCPC), Champion**.

Skills

- Programming **C/C++, Python, CUDA**.
- Systems **MPI, Tensorflow, Pytorch, NCCL**.

Talks

- Nov 2021 *Towards Efficient Training for Large-scale Deep Learning Models*, **AWS AI Lab**.
- Apr 2021 *DGCL: An Efficient Communication Library for Distributed GNN Training*, **Eurosys 2021**.

Projects

Systems for large-scale machine learning

TensorOpt: Training large-scale DNNs with auto-parallel.

- Supports distributed training of large models (e.g., Transformer, WideResNet) using limited GPU memory
- Optimally decides the parallelization strategy and automatically generates code for operators in a DNN
- Developed on top of TensorFlow with user-friendly Python APIs

EDL: An Elastic Deep Learning System on GPUs.

- Supports elastic deep learning, i.e., dynamically adjust the number of GPU at runtime
- Stop-free scaling and dynamic data pipeline on Horovod

FlexPS: A Parameter Server with Flexible Parallelism Control.

- A novel multi-stage abstraction to support flexible parallelism control in parameter server
- Optimizations to reduce the overhead of parallelism adjustment

Systems for graph neural networks

DGCL: A Distributed Graph Communication Library for GNN systems.

- A general library to scale single-GPU GNN systems (e.g., DGL and PyG) to the multi-GPU setting
- Efficient communication kernels optimizing for load balancing and bandwidth utilization on NVLinks, PCIe and InfiniBand

Seastar: A Vertex-centric GNN System.

- A vertex-centric programming model for GNNs
- Kernel optimizations such as operator fusion and vertex parallelism to reduce GPU memory consumption and improve training efficiency

Large-scale cluster scheduling

PPS: Fair And Efficient Scheduling for Multi-Tenant GPU Clusters.

- Probabilistic prediction based scheduling for clusters with thousands of GPUs
- Black-box and non-preemptive scheduling for GPU jobs
- Achieves both efficiency and fairness at the same time

Ursa: A Framework for both Resource Scheduling and Job Execution.

- Captures dynamic resource needs at runtime and enables fine-grained, timely scheduling
- Achieves high resource utilization, which translates into shorter makespan and average JCT

Publications

- [1] *TensorOpt: Exploring the Tradeoffs in Distributed DNN Training with Auto-Parallelism*, **TPDS 2022**,
Z. Cai, K. Ma, X. Yan, Y. Wu, Y. Huang, J. Cheng, T. Su, F. Yu.
- [2] *PPS: Fair and Efficient Scheduling for Multi-Tenant GPU Clusters*, **NSDI 2022** (Under Review),
Z. Cai, K. Ma, X. Yan, Y. Zhang, Z. Liu, Y. Feng, C. Li, J. Cheng.
- [3] *DGCL: An Efficient Communication Library for Distributed GNN Training*, **Eurosys 2021**,
Z. Cai, X. Yan, Y. Wu, K. Ma, J. Cheng, F. Yu.
- [4] *Seastar: Vertex-Centric Programming for Graph Neural Networks*, **Eurosys 2021**,
Y. Wu, K. Ma, **Z. Cai**, T. Jin, B. Li, C. Zheng, J. Cheng, F. Yu.
- [5] *Elastic Deep Learning in Multi-Tenant GPU Clusters*, **TPDS 2021**,
Y. Wu, K. Ma, X. Yan, Z. Liu, **Z. Cai**, Y. Huang, J. Cheng, H. Yuan, F. Yu.
- [6] *Improving Resource Utilization by Timely Fine-Grained Scheduling*, **Eurosys 2020**,
T. Jin, **Z. Cai**, B. Li, C. Zheng, G. Jiang, J. Cheng.
- [7] *FlexPS: Flexible Parallelism Control in Parameter Server Architecture*, **VLDB 2018**,
Y. Huang, T. Jin, Y. Wu, **Z. Cai**, X. Yan, Y. Guo, F. Yang, J. Li, and J. Cheng.
- [8] *Scalable De Novo Genome Assembly Using Pregel*, **ICDE 2018**,
D. Yan, H. Chen, **Z. Cai**, J. Cheng, and B. Shao.